

Maestro Cloud Control

Enabling Access to GCP Billing

Admin Guide

M3AG-3

February 2026

Version 1.3

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PREFACE

ABOUT THIS GUIDE

The guide gives detailed instructions on what steps should be taken to enable programmatic access to GCP Account billing and how to allow managing virtual resources via Maestro tools.

These steps include enabling billing data collection and enabling management .

AUDIENCE

The guide is designed for administrators of Google Cloud Platform accounts who want to enable access to their billing information and infrastructure management for the Maestro application.

PRE-REQUISITES

Before you can perform the steps described in the guide, ensure that these conditions are met:

1. You have an active GCP account.
2. You have access to the account with the owner user permissions.
3. You know the ID of the account to which the access will be provided.

Also make sure that the following APIs are enabled on the target account (replace the values in UPPER_CASE with your real Google Project IDs)

- **Cloud Deployment Manager V2 API:**
https://console.developers.google.com/apis/api/deploymentmanager.googleapis.com/overview?project=MANAGEMENT_PROJECT_ID
- **Compute Engine API:**
https://console.cloud.google.com/apis/library/compute.googleapis.com?project=MANAGEMENT_PROJECT_ID
- **BigQuery API:**
https://console.cloud.google.com/apis/library/bigquery.googleapis.com?project=BILLING_PROJECT_ID

EXPECTED RESULTS

Once all the steps have been completed, you will have the following information to be shared with Maestro team:

- The name or names of the service accounts to be used by Maestro to access your Google Cloud infrastructure.
- The private key or keys for each of the service accounts.
- BigQuery Dataset name used for billing data export.
- BigQuery Dataset name used for detailed usage costs gathering.

1 ENABLING ACCESS TO GCP PROJETS

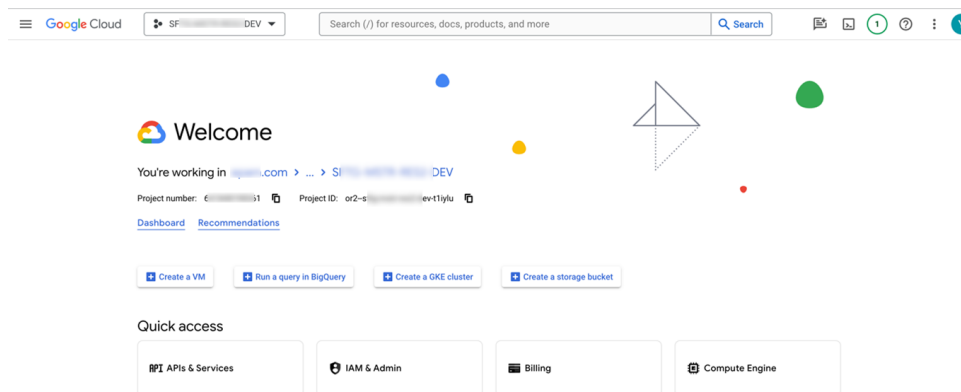
The permissions Maestro needs for working with infrastructures in GCP, can be generally split by three aims: billing, management, and terraform. You can create a single or separate service accounts for enabling necessary permissions. The choice depends on your preferences or corporate policies.

2 ENABLING MANAGEMENT CONFIGURATION

2.1 CREATING A SERVICE ACCOUNT

To create the service account:

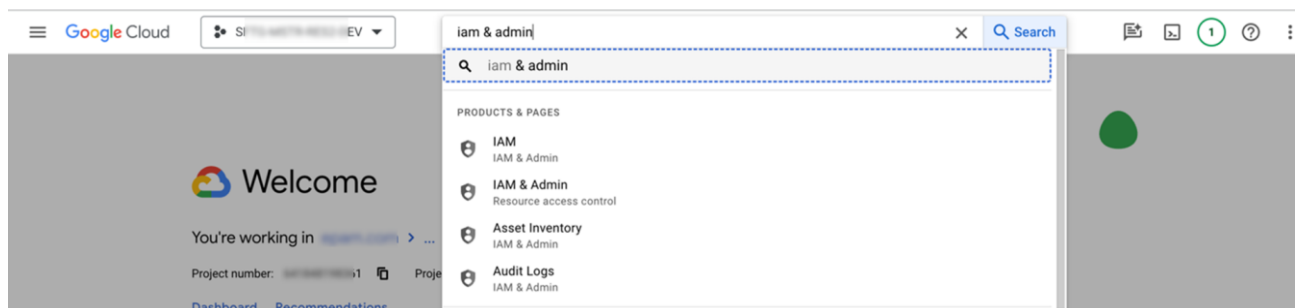
1. Login to the Google account and choose the project:



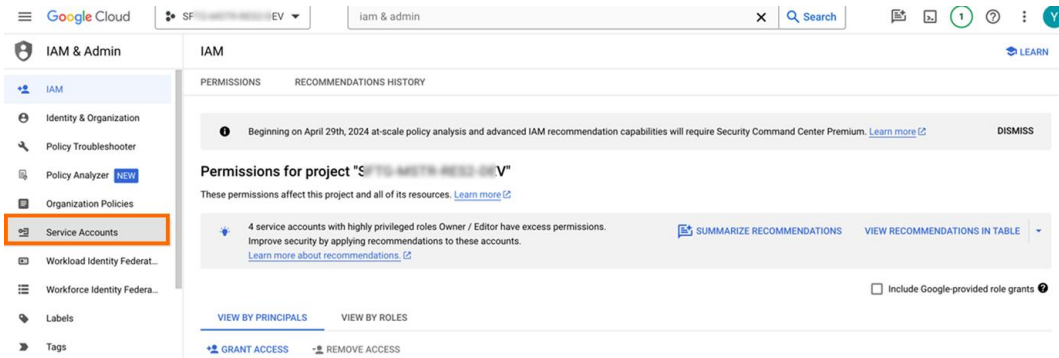
2. Search and open “iam & admin”.



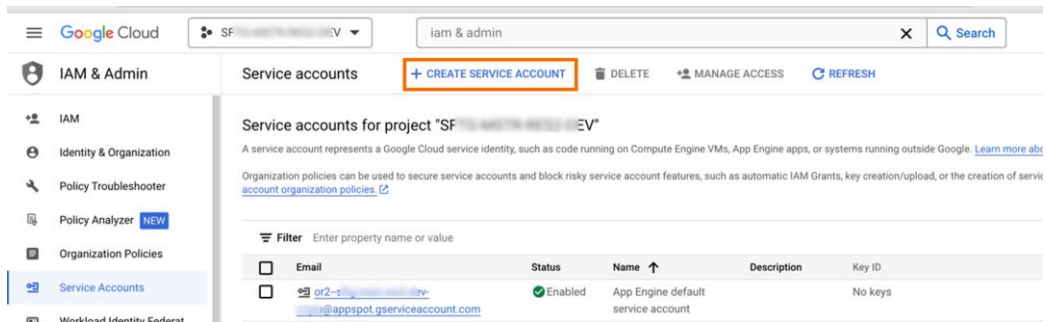
Administrator who will execute the next step needs “Service Account Admin” permissions



3. Open “Service Accounts”



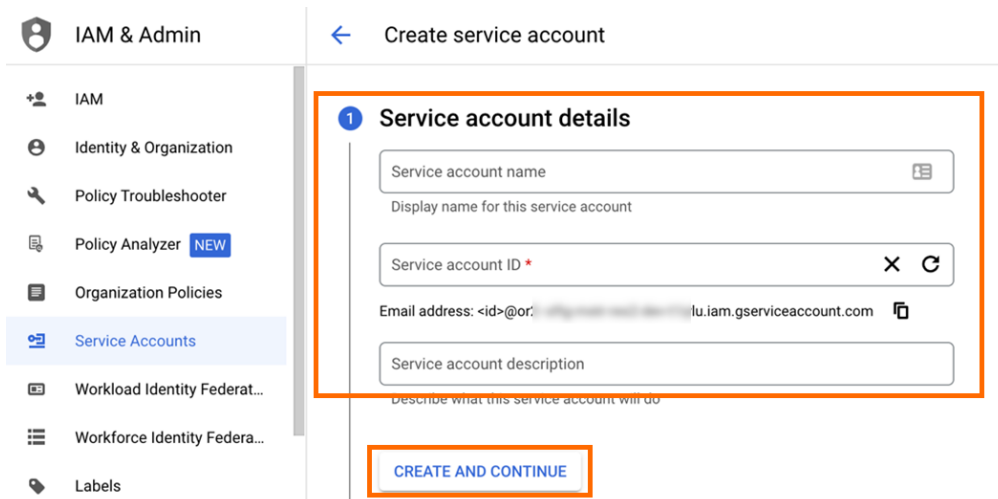
4. Choose “Create service account”



5. Fill the required **Service account details** fields:

- Service account name
- Service account ID
- Service account description(optional)

And click “Create and continue”



6. Grant permissions.

Chose the required role for management recourses. You can either combine all permissions under a single account, or create different accounts for each purpose:

Management:

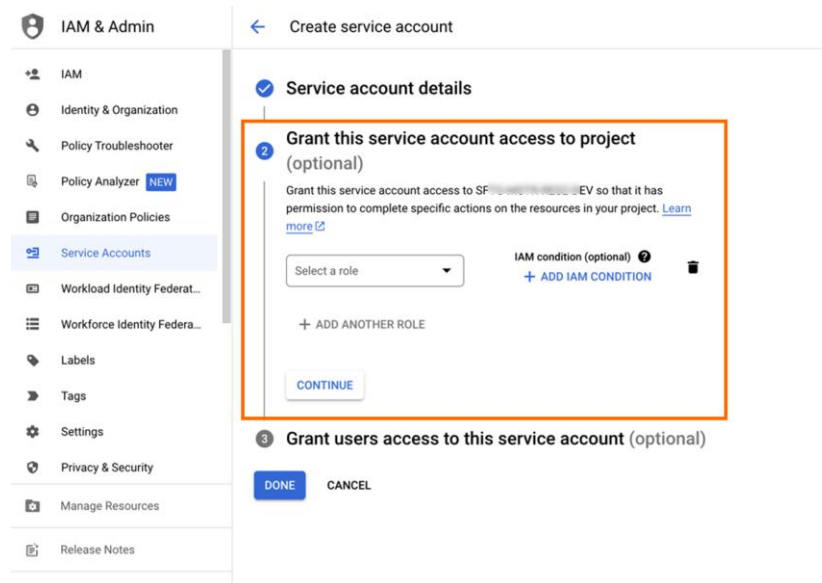
- Compute Admin
- Deployment Manager Editor
- DNS Administrator
- Kubernetes Engine Viewer
- Monitoring Metric Writer
- Service Account User

Billing:

- BigQuery Data viewer
- BigQuery Data Editor
- BigQuery Job User

Terraform: the permissions depend on the scope you want to cover.

Once all the roles are added, click **“Continue”**



7. Grant users access to this service account (optional)

Granting users access to this service account is not required for management, you can leave it empty.

Click **“Done”**

Maestro – Enabling Access to GCP Billing

IAM & Admin

- IAM
- Identity & Organization
- Policy Troubleshooter
- Policy Analyzer **NEW**
- Organization Policies
- Service Accounts**
- Workload Identity Federat...
- Workforce Identity Federa...
- Labels
- Tags
- Settings
- Privacy & Security
- Manage Resources

Create service account

✓ **Service account details**

✓ **Grant this service account access to project (optional)**

3 **Grant users access to this service account (optional)**

Grant access to users or groups that need to perform actions as this service account. [Learn more](#)

Service account users role ?

Grant users the permissions to deploy jobs and VMs with this service account

Service account admins role ?

Grant users the permission to administer this service account

DONE CANCEL

8. Find the service account that has been created in the previous steps.

IAM & Admin

Service accounts + CREATE SERVICE ACCOUNT DELETE MANAGE ACCESS REFRESH [LEARN](#)

A service account represents a Google Cloud service identity, such as code running on Compute Engine VMs, App Engine apps, or systems running outside Google. [Learn more about service accounts](#)

Organization policies can be used to secure service accounts and block risky service account features, such as automatic IAM Grants, key creation/upload, or the creation of service accounts entirely. [Learn more about service account organization policies](#)

Filter Enter property name or value

Email	Status	Name	Description	Key ID	Key creation di	Actions
gc1-af8p3eas-dev-iam-gcp@agppspot.gserviceaccount.com	Enabled	gc1-af8p3eas-dev-iam-gcp	Agg...	No keys		
6-...@compute@developer.gserviceaccount.com	Enabled	6-...	Co...	No keys		
maestro3-dev-iam-gcp@maestro3-dev-iam.iam.gserviceaccount.com	Enabled	maestro3-dev-iam-gcp	service account for management	No keys		

9. On the right, open **Actions-> manage keys**

IAM & Admin

Service accounts + CREATE SERVICE ACCOUNT DELETE MANAGE ACCESS REFRESH [LEARN](#)

A service account represents a Google Cloud service identity, such as code running on Compute Engine VMs, App Engine apps, or systems running outside Google. [Learn more about service accounts](#)

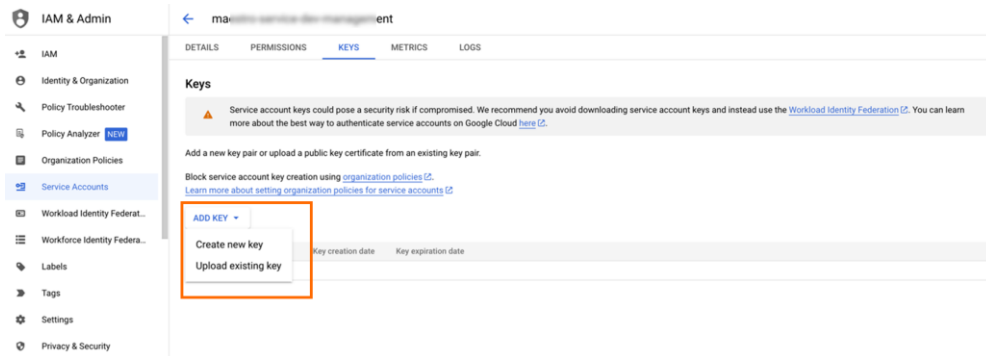
Organization policies can be used to secure service accounts and block risky service account features, such as automatic IAM Grants, key creation/upload, or the creation of service accounts entirely. [Learn more about service account organization policies](#)

Filter Enter property name or value

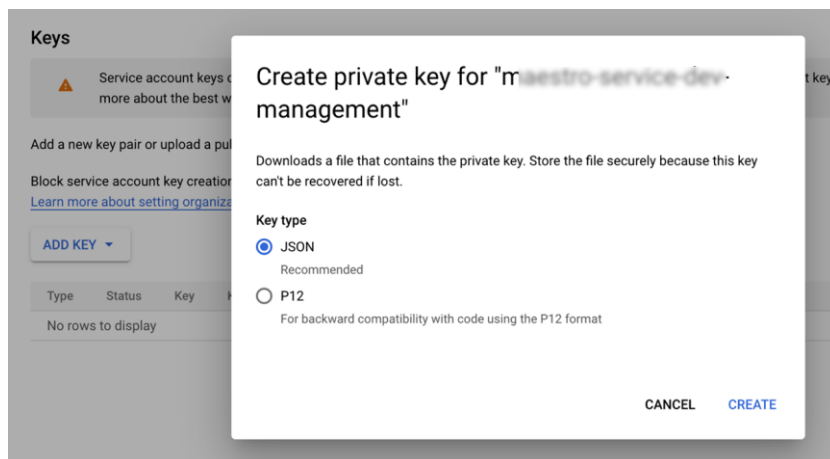
Email	Status	Name	Description	Key ID	Key creation di	Actions
gc1-af8p3eas-dev-iam-gcp@agppspot.gserviceaccount.com	Enabled	gc1-af8p3eas-dev-iam-gcp	Agg...	No keys		
6-...@compute@developer.gserviceaccount.com	Enabled	6-...	Co...	No keys		
maestro3-dev-iam-gcp@maestro3-dev-iam.iam.gserviceaccount.com	Enabled	maestro3-dev-iam-gcp	service account for management	No keys		
maestro3-dev-iam-gcp@maestro3-dev-iam.iam.gserviceaccount.com	Enabled	maestro3-dev-iam-gcp	maestro3 dev terraform onprem	cf...		
maestro3-dev-iam-gcp@maestro3-dev-iam.iam.gserviceaccount.com	Enabled	maestro3-dev-iam-gcp	service account for dev env	ec...		
maestro3-dev-iam-gcp@maestro3-dev-iam.iam.gserviceaccount.com	Enabled	maestro3-dev-iam-gcp	maestro3 dev onprem management	D5bC...		

Manage details
Manage permissions
Manage keys
View metrics
View logs
Disable
Delete

10. Keys -> ADD KEY-> Create new key



Choose key type: json



Save the key and send it to the Maestro team.

2.2 ENABLING BILLING CONFIGURATION

Permissions required for the configuration.

To enable and configure the export of Google Cloud billing *usage cost* data to a BigQuery dataset, you need the following permissions:

- For Cloud Billing, you need either the Billing Account Costs Manager role or the Billing Account Administrator role on the target Cloud Billing account.
- For BigQuery, you need the [BigQuery User role for the Google Cloud project](#) that contains the BigQuery dataset to be used to store the Cloud Billing data.

To enable and configure the export of Cloud Billing *pricing* data, you need the following permissions:

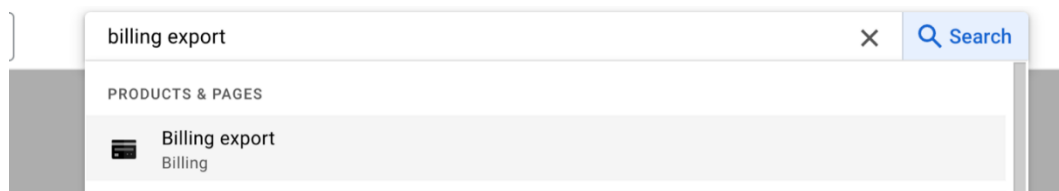
- For Cloud Billing, you need the Billing Account Administrator role on the target Cloud Billing account.
- For BigQuery, you need the [BigQuery Admin role for the Google Cloud project](#) that contains the BigQuery dataset to be used to store the Cloud Billing pricing data.
- For the Google Cloud project containing the target dataset, you need the `resourcemanager.projects.update` permission. This permission is included in the roles/editor role.

2.2.1 Export Cloud Billing Data to BigQuery

Cloud Billing export to BigQuery enables you to export detailed Google Cloud billing data (such as usage, cost estimates, and pricing data) automatically throughout the day to a BigQuery dataset that you specify. Then you can access your Cloud Billing data from BigQuery for detailed analysis, or use a tool like Looker Studio to visualize your data. You can also use this export method to export data to a JSON file.

You need to configure export for each required project according to the steps given below:

1. Find and open the **“Billing export”** configurations in your account.

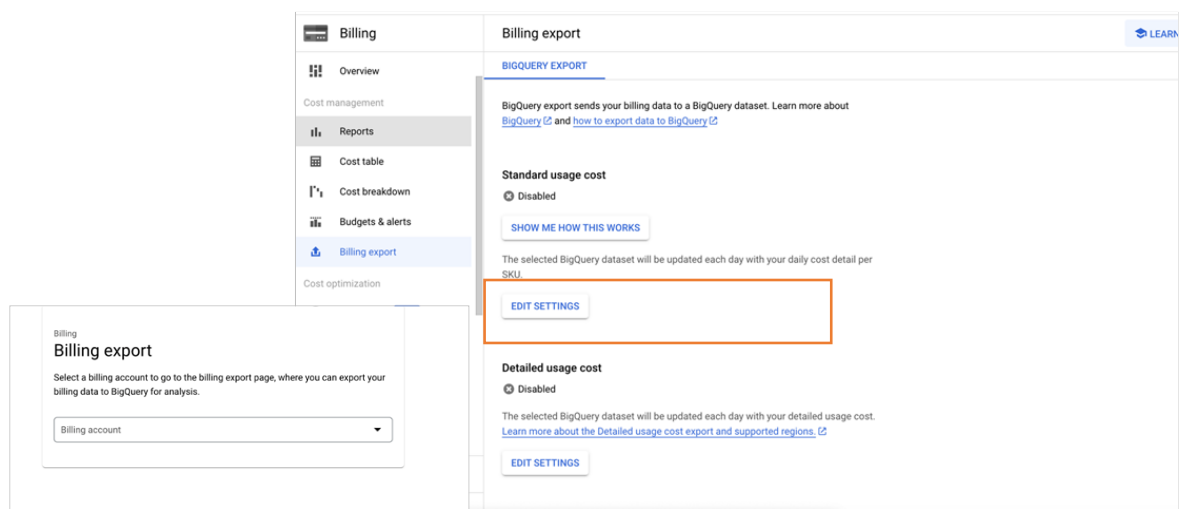


2. Select a billing account to go to the billing export page, where you can export your billing data to BigQuery for analysis.

When you enable the standard usage cost data export for the first time in Cloud Billing, if you select a dataset configured to use a multi-region location (EU or US), Cloud Billing data will be available retroactively from the start of the previous month. The data is exported in chronological order. For the initial backfill of exported data, it might take up to 5 days for your retroactive Cloud Billing data to finish exporting before you start seeing your most recent usage data.

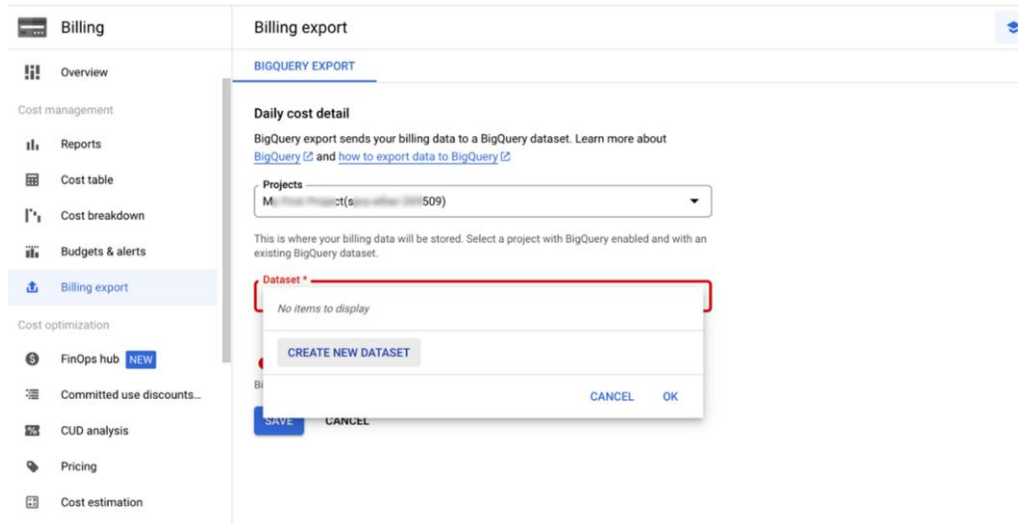
In your BigQuery dataset, your standard Google Cloud usage cost data is loaded into a data table named **gcp_billing_export_v1_<BILLING_ACCOUNT_ID>**.

Click **Edit settings** for the **“Standard usage cost”**



3. Configure the Standard usage cost:

- Select project
- Click “Create new dataset”
- Set a name for dataset
- Save new settings.

**Send dataset name to the Maestro team**

BigQuery export will create a table with your billing data in the selected dataset.

2.2.2 Configure the Detailed Usage Cost

The *detailed usage cost data* provides all of the information included in the [standard usage cost data](#), along with additional fields that provide granular, resource-level cost data, like a virtual machine or SSD that generates usage. The detailed export includes granular cost information about the following services:

- Compute Engine
- Cloud Storage
- Google Kubernetes Engine (GKE)

To view a breakdown of GKE cluster costs in a detailed data export, you must *also* [enable cost allocation for GKE](#).

- Cloud Functions
- Cloud Run
- Cloud SQL
- Spanner
- App Engine
- Firestore and Datastore
- Memorystore for Redis

In your BigQuery dataset, your detailed Google Cloud usage cost data is loaded into a data table named `gcp_billing_export_resource_v1_<BILLING_ACCOUNT_ID>`.

For Detailed usage cost, click **Edit settings:**

Detailed usage cost

 Disabled

The selected BigQuery dataset will be updated each day with your detailed usage cost.

[Learn more about the Detailed usage cost export and supported regions.](#)

[EDIT SETTINGS](#)

In the settings:

- Choose project and
- Click “Create new dataset”
- Enter dataset name.
- Save new settings.

 Billing

 Overview

Cost management

 Reports

 Cost table

 Cost breakdown

 Budgets & alerts

 Billing export

Cost optimization

 FinOps hub NEW

 Committed use discounts...

 CUD analysis

 Pricing

Billing export

BIGQUERY EXPORT

Detailed usage cost

The selected BigQuery dataset will be updated each day with your detailed usage cost.
[Learn more about the Detailed usage cost export and supported regions.](#)

Projects

M...t(509)

This is where your billing data will be stored. Select a project with BigQuery enabled and with an existing BigQuery dataset.

Dataset *

No items to display

CREATE NEW DATASET

CANCEL OK

Save the resulting dataset name for sharing it with Maestro

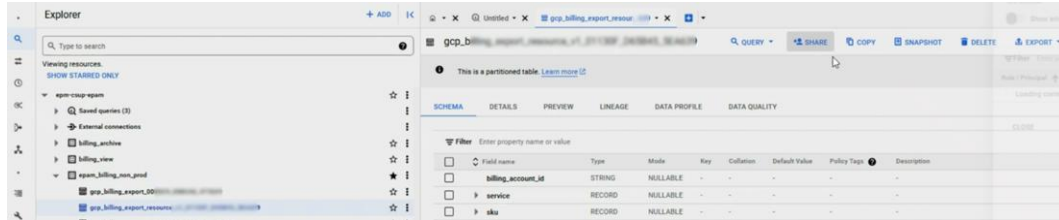


Google Cloud products report usage and cost data to Cloud Billing processes at varying intervals. As a result, you might see a delay between your use of Google Cloud services, and the usage and costs being available to view in Cloud Billing. Typically, your costs are available within a day, but can sometimes take more than 24 hours.

2.2.3 Configure Permissions

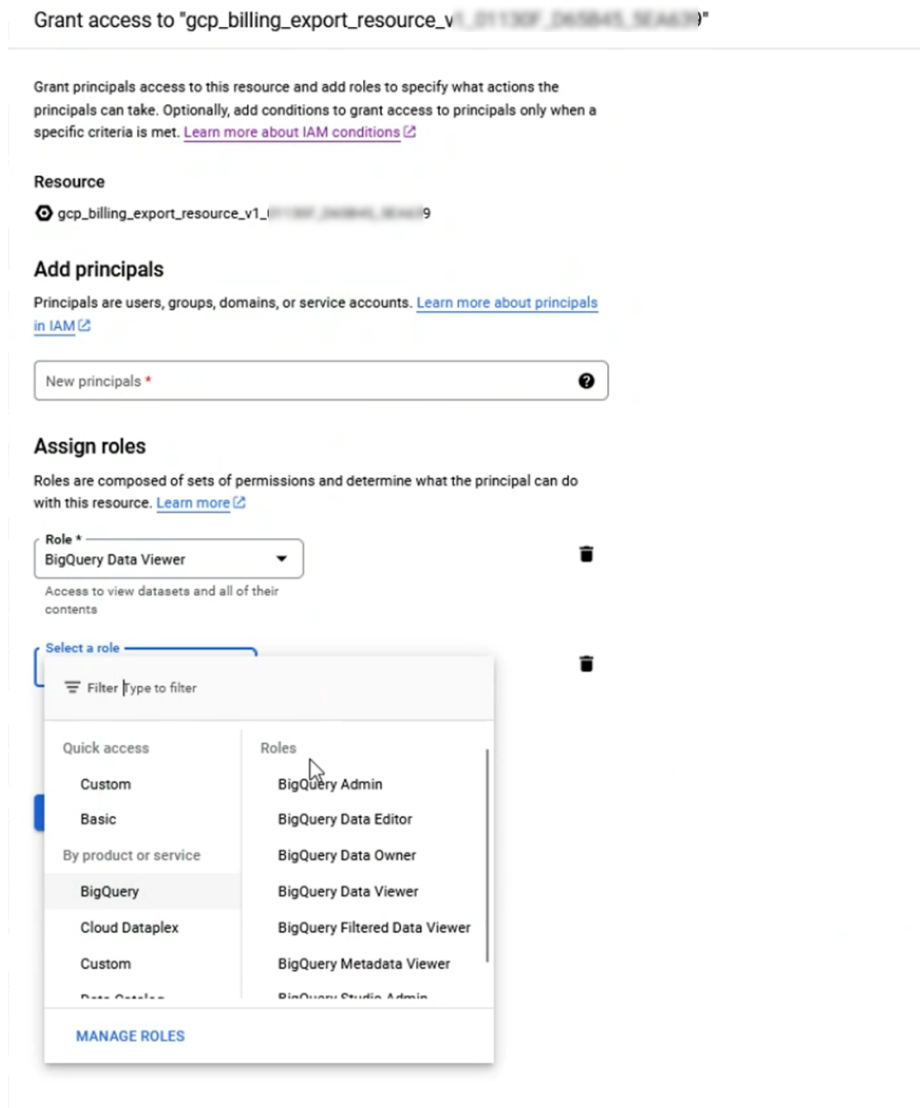
The steps below should be made for both datasets.

1. Choose a dataset and in the top menu click “SHARE.”



In the opened window:

- Find your service user.
- Select the “BigQuery Data viewer” and “BigQuery Data Editor” roles.

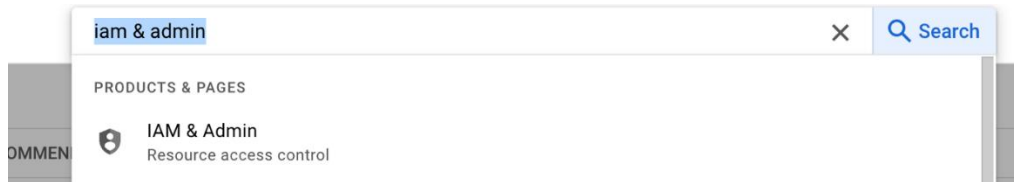


Possible issue:

Unfortunately, time to time users faced problems when try adding “BigQuery Data Editor” role at this step.

Solution:

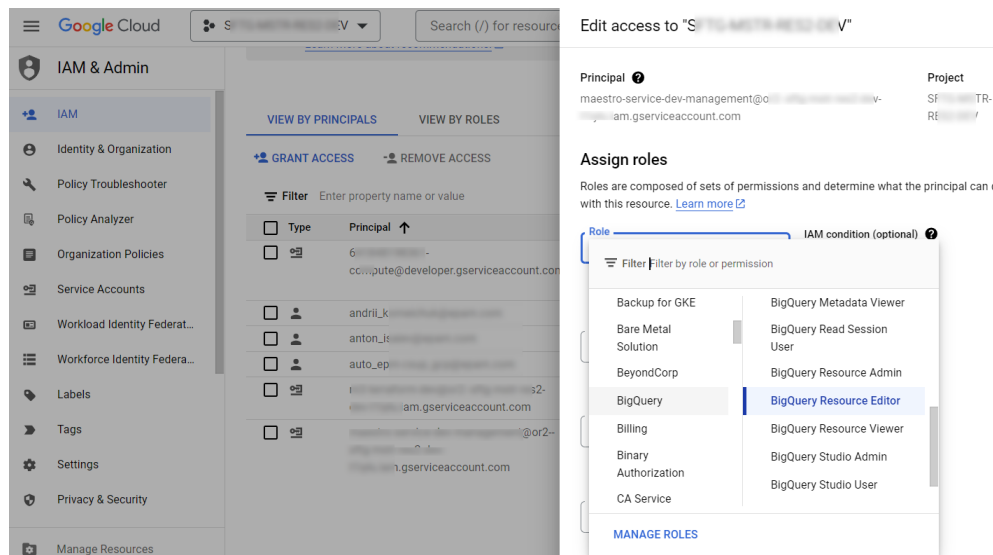
1. Go to “IAM & Admin” menu



2. Choose a service account, which you will be using for billing.



3. Find BigQuery -> BigQuery data editor.



3 RELATED ARTIFACTS LIST

The successful setup of GCP projects for Maestro access needs the following materials:

Artifact Name	Description	Provided by
gcp-tenant.sh	The script to connect a GCP project to Maestro	Maestro team
gcp.env	The environment variables file required for correct script execution	Maestro team
SECURITY_GROUPS_FILE	A .json file storing the security groups configuration for the target GCP project	

VERSION HISTORY

Version	Date	Summary
1.3	February 23, 2026	Reviewed
1.2	April 24, 2024	Necessary APIs descriptions added, permission requirements clarified
1.1	April 2, 2024	Reviewed and updated
1.0	June 12, 2023	Initial version published